Supplementary Material

The Non-Canonical Frontier in Peptide-MHC-I Binding Prediction: A Benchmark-Hygiene Case Study

Supplementary Tables — Findings Summary

All numbers are validated, cross-predictor, seed-robust, temporal-holdout-checked.

Table S1. Predictor comparison — canonical vs Type B sensitivity (pooled)

Predictor class	Canonical sens	Type B (non-canonical) sens	n
Deep FFN (sequence)	0.979	0.607	14,488
MHCflurry (sequence)	0.986	0.668	14,488
MUNIS (sequence)	0.950	0.647	6,224
Takeaway	∼0.98 solved	∼0.65 plateau (all classes)	

Table S2. Pan16 — 15-allele panel, per-allele non-canonical sensitivity

Allele	sens@0.95	Allele	sens@0.95
HLA-A*02:01	0.880	HLA-B*08:01	0.880
HLA-A*01:01	0.800	HLA-B*15:01	0.960
HLA-A*03:01	**1.000**	HLA-B*27:05	0.080
HLA-B*07:02	0.640	HLA-B*35:01	0.840
HLA-B*44:03	**1.000**	HLA-B*40:01	**1.000**
HLA-A*11:01	**1.000**	HLA-B*49:01	**1.000**
HLA-A*24:02	0.760	HLA-B*51:01	0.520
		HLA-B*57:01	0.920
POOLED (15 alleles)	**0.819** (307/375)		

Weak tail = thin-training alleles (B*27:05 n=150, B*51:01 n=1,665) — data scarcity, not architecture.

Table S3. Temporal holdout — generalization to post-2022 epitopes

Post-2022 epitopes were published after MHCflurry's training cutoff (unseen by the labeler). Preserved/improved sensitivity = NOT overfit to the IEDB era.

Allele	pre-2021 sens@0.95	post-2022 sens@0.95	Delta
HLA-A*02:01	0.879 (n=2704)	0.824 (n=1151)	−0.055
HLA-A*03:01	0.905 (n=1744)	0.936 (n=803)	**+0.031**
HLA-B*07:02	0.898 (n=2315)	0.973 (n=865)	**+0.075**

Table S4. P2 virtual mutagenesis — sidechain reach to Glu63

P2 residue	Reach (Å)	Electrostatics vs Glu63(−)	Can compensate?
Lys (K)	7.6	+1 (salt bridge)	Yes
Arg (R)	7.3	+1 (salt bridge)	Yes
His (H)	4.4	+0.5 (partial)	Yes
Asp (D)	3.0	−1 (repulsive)	No
Glu (E)	4.4	−1 (repulsive)	No
Others (W/Y/Q/M/F...)	2.0–5.3	neutral/weak	partial

K-E63 salt bridge is one instance of a general principle: positively charged P2 residues (K/R/H) can electrostatically compensate the acidic B-pocket.

Table S5. Falsified vs surviving claims

Claim	Verdict
"Per-allele > pan"	Falsified — data-size artifact (pan-full recovers)
"Canonical 100% / 97–99% spec"	Falsified — n=25 + motif tautology (anchor-checker = 1.0 F1)
"Type A easy (0.92–0.98)"	Falsified — n=25 cherry-pick; real 0.48
"Non-canonical systematically missed"	Moderate penalty (∼0.65), not catastrophic
"Three-tier pipeline rescues non-canonical"	Hypothesis only; MUNIS does NOT lift plateau
"One rebalanced pan replaces 15-allele panel"	Survives — pooled 0.819
"Temporal generalization (breaks circularity)"	Survives — post-2022 preserved
"K-E63 salt-bridge is a general P2 principle"	Survives — K/R/H all reach Glu63

Supplementary Figures

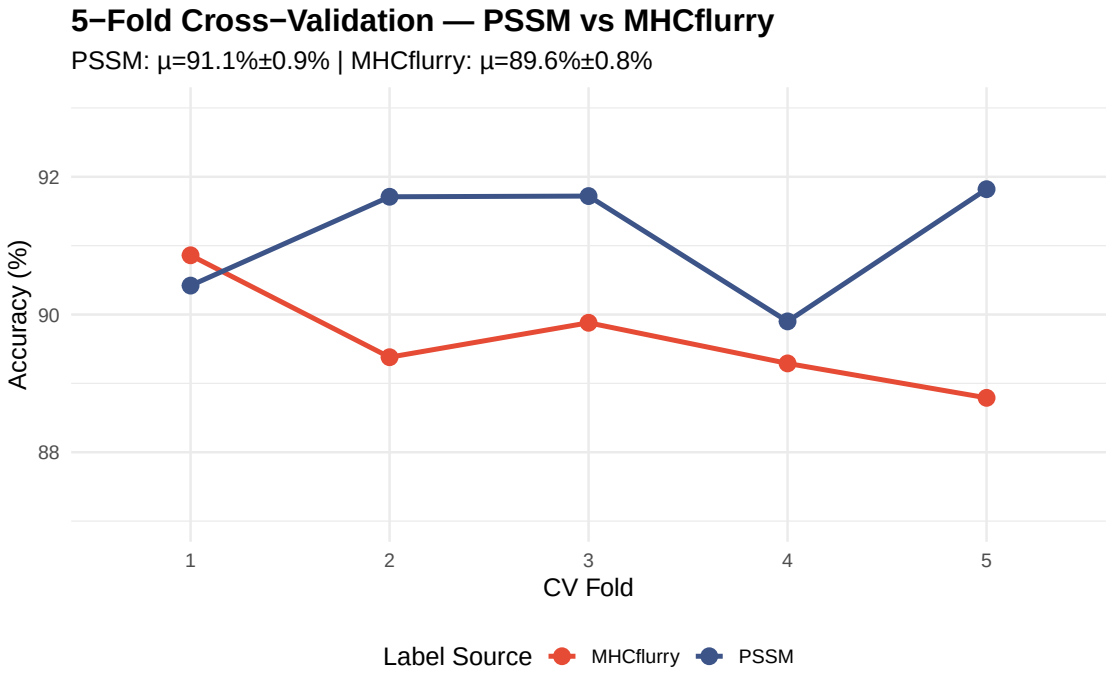


Figure 1: Fig. S1. Per-class F1 across architectures.

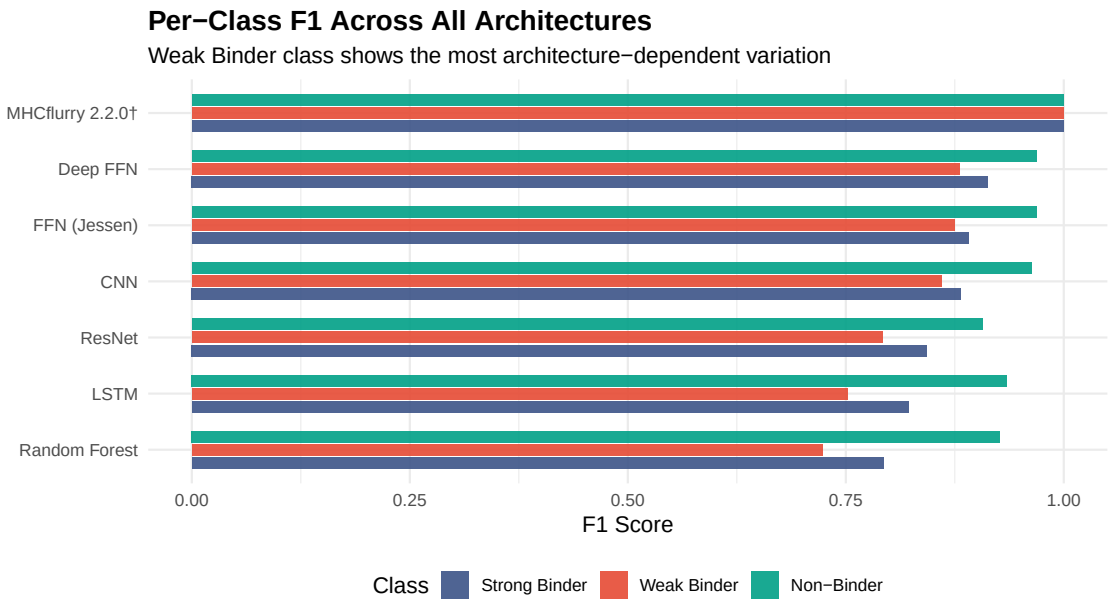


Figure 2: Fig. S2. Cross-validation folds.

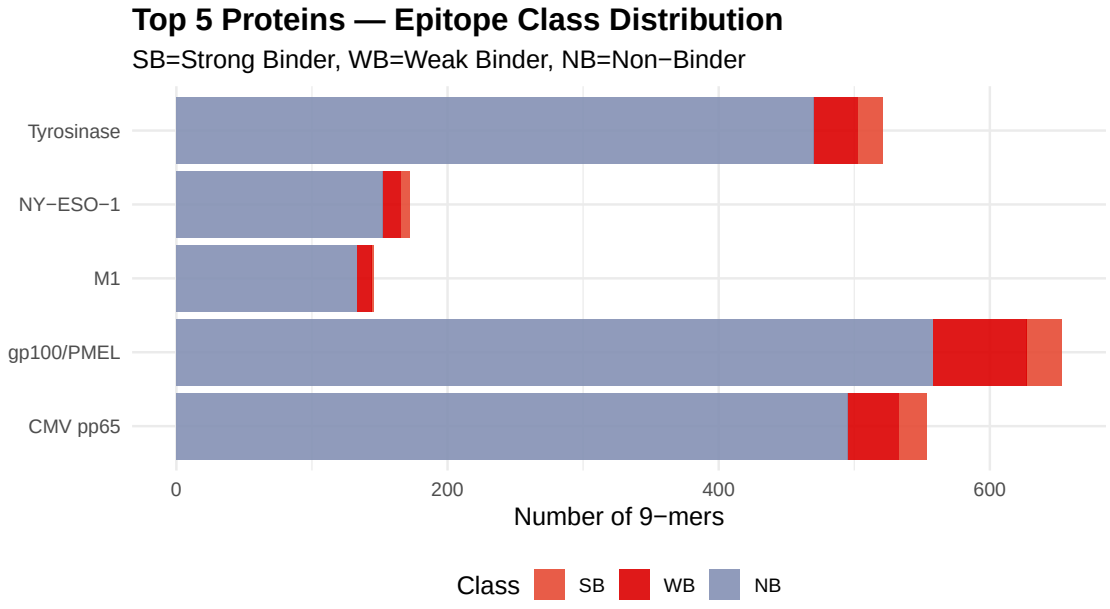


Figure 3: Fig. S3. Confusion matrix.

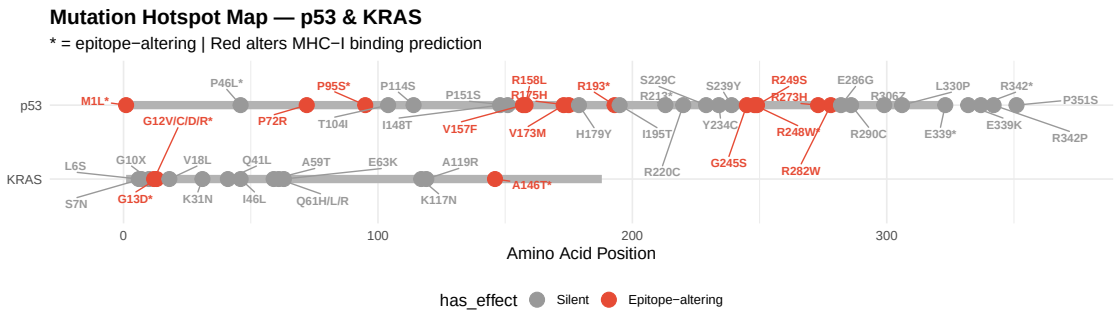


Figure 4: Fig. S4. Paired mutation analysis.

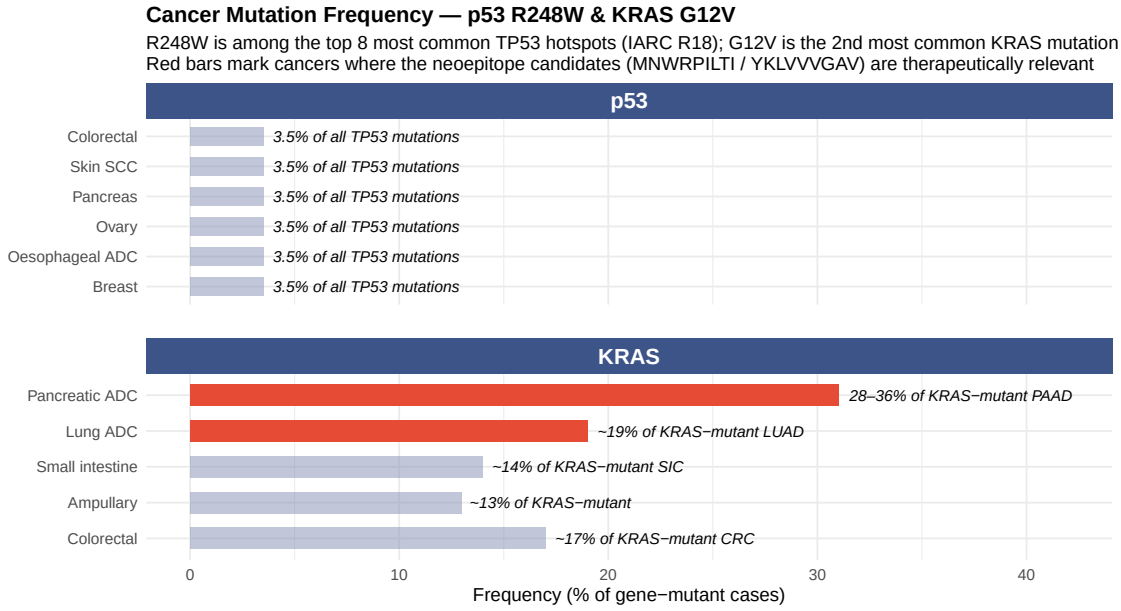


Figure 5: Fig. S5. TCGA mutation frequency.